10 Chapter

Numerical Solution of Ordinary Differential Equations

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10.1 INTRODUCTION

The methods of solution so far presented are applicable to a limited class of differential equations. Frequently differential equations appearing in physical problems do not belong to any of these familiar types and one is obliged to resort to numerical methods. These methods are of even greater importance when we realize that computing machines are now available which reduce numerical work considerably.

A number of numerical methods are available for the solution to first order differential equations of the form:

$$\frac{dy}{dx} = f(x, y), \text{ given } y(x_0) = y_0. \qquad \dots (i)$$

These methods yield solutions either as a power series in x from which the values of y can be found by direct substitution, or as a set of values of x and y. The methods of Picard and Taylor series belong to the former class of solution whereas those of Euler, Runge-Kutta, Milne Adams-Bashforth, etc., belong to the latter class. In these later methods, the values of y are calculated in short steps for equal intervals of x and are therefore, termed as step-by-step methods.

Euler and Runge-Kutta methods are used for computing y over a limited range of x-values where as Milne and Adams-Bashforth methods may be applied for finding y over a wider range of x values. These later methods require starting values which are found by Picard's or Taylor series or Runge-Kutta methods.

The initial condition in (*i*) is specified at the point (x_0). Such problems in which all the initial conditions are given at the initial point only are called *initial value problems*. But there are problems involving second and higher order differential equations in which the conditions may be given at two or more points. These are known as *boundary value problems*. Now we shall discuss the various methods to find the numerical solution to the ordinary differential equations.

10.2 TAYLOR'S SERIES METHOD

Let $y = f(x)$ be a solution to the equation

$$\frac{dy}{dx} = f(x, y)$$

with

$$y(x_0) = y_0 \qquad \text{... (i)}$$

Expanding it by Taylor's series about the point x_0, we get

$$f(x) = f(x_0) + \frac{(x - x_0)}{1!} f'(x_0) + \frac{(x - x_0)^2}{2!} + f''(x_0) + \ldots$$

this may be written as

$$y = f(x) = y_0 + \frac{(x - x_0)}{1!} y'_0 + \frac{(x - x_0)^2}{2!} y''_0 + \ldots$$

putting $x = x_1 = x_0 + h$ we get

$$f(x_1) = y_1 = y_0 + \frac{h}{1!} y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \ldots \qquad \text{... (ii)}$$

Similarly we obtain

$$y_{n+1} = y_n + \frac{h}{1!} y'_n + \frac{h^2}{2!} y''_n + \frac{h^3}{3!} y'''_n + \ldots \qquad \text{... (iii)}$$

equation (*iii*) may be written as

$$y_{n+1} = y_n + \frac{h}{1!} y'_n + \frac{h^2}{2!} y''_n + O(h^3) \qquad \text{... (iv)}$$

where $O(h^3)$ means that all the succeeding terms contain the third and higher powers of h. If the terms containing the third and higher powers of h are neglected then the local truncation error in the solution is kh^3 where k is a constant. For a better approximation terms containing higher powers of h are considered.

Note: Taylor's series method is applicable only when the various derivatives of $f(x, y)$ exist and the value of $(x - x_0)$ in the expansion of $y = f(x)$ near x_0 must be very small so that the series converge.

SOLVED EXAMPLES

Solve for x (1.2)

Example 10.1. Solve $\frac{dy}{dx} = x + y$, $y(1) = 0$, numerically upto $x = 0.2$ with $h = 0.1$.

Solution: We have $x_0 = 1, y_0 = 0$

$$\frac{dy}{dx} = y' = x + y \;\Rightarrow\; y'_0 = 1 + 0 = 1,$$

$y' = x + y \Rightarrow y' = 1 + y$

$$\frac{d^2y}{dx^2} = y'' = 1 + y' \Rightarrow y_0'' = 1 + 1 = 2$$

$$\frac{d^3y}{dx^3} = y''' = y'' \Rightarrow y_0''' = 2$$

$$\frac{d^4y}{dx^4} = y^{iv} = y''' \Rightarrow y^{iv} = 2$$

Substituting the above values in

$$y_1 = y_0 + \frac{h}{1!}y_0' + \frac{h^2}{2!}y_0'' + \frac{h^3}{3!}y_0''' + \frac{h^4}{4!}y_0^{iv} + \frac{h^5}{5!}y_0^{v} + \ldots \qquad \ldots(i)$$

$$y_1 = 0 + (0.1) + \frac{(0.1)^2}{2}.2 + \frac{(0.1)^3}{6}.2 + \frac{(0.1)^4}{24}.2 + \ldots$$

$\Rightarrow$ $y_1 = 0.11033847$

$\therefore$ $y_1 = y(0.1) \approx 0.110.$

Now $x_1 = x_0 + h = 1 + 0.1 = 1.1$

$$y_1' = x_1 + y_1 = 1.1 + 0.110 = 1.21$$

$$y_1'' = 1 + y = 1 + 1.21 = 2.21$$

$$y_1''' = y_1'' = 2.21,$$

$$y_1^{iv} = 2.21$$

Substituting the above values in (*i*), we get

$$y_2 = 0.110 + (0.1)(1.21) + \frac{(0.1)^2}{2}(2.21) + \frac{(0.1)^3}{6}(2.21) + \frac{(0.1)^4}{24}(2.21) + \ldots$$

$\therefore$ $y_2 = 0.24205$

$\therefore$ $y(0.2) = 0.242.$

Example 10.2. **Given $\frac{dy}{dx} = 1 + xy$ with initial condition that $y = 1, x = 0$, compute $y(0.1)$ correct to four places of decimal by using Taylor's series method.**

Solution: $\frac{dy}{dx} = 1 + xy$ and $y(0) = 1$

$\therefore$ $y'(0) = 1 + 0 \times 1 = 1$

Differentiating the given equation w.r.t., x we get

$$\frac{d^2y}{dx^2} = y + x\frac{dy}{dx}$$

$u''(x) = -8(u')^2 - 8uu'' - 4(u')^2 - 4uu'' + x(-8u'u'' - 4u'u'' - 4uu''')$

$u^{iv}(x_0) = -8(0)^2 - 8(1)(-1) - 4(1)(-1) + 0 = 12$

Substituting these values in the Taylor's series, we get

$$u(0.1) = 1 + (0.1-0) \times 0 + \frac{(0.1-0)^2}{2!}(-1) + \frac{(0.1-0)^3}{3!} \times 0 + \frac{(0.1-0)^4}{4!} \times 12$$

$$= 0.99505$$

10.3 PICARD'S METHOD

Let us want to solve $\frac{dy}{dx} = f(x, y)$

with initial condition $y = y_0$ when $x = x_0$ ... (i)

Integrating (1) between the limit x_0 to x, we get

$$\int_{y_0}^{y} dy = \int_{x_0}^{x} f(x,y)\,dx$$

$$y - y_0 = \int_{x_0}^{x} f(x,y)\,dx$$

$$y = y_0 + \int_{x_0}^{x} f(x,y)\,dx \quad \text{... (ii)}$$

Now when we replace $y = y_1$ in $f(x, y)$, let us get first approximation of solution say y_1, thus

$$y_1 = y_0 + \int_{x_0}^{x} f(x, j_0)\,dx$$

On solving we get y_1 and then continue in this way, we get

$$y_2 = y_0 + \int_{x_0}^{x} f(x,y)\,dx$$

....

....

$$y_n = y_0 + \int_{x_0}^{x} f(x, y_{n-1})\,dx$$

Remark: Picard's method has some limitations, Picard's method is not useful for computer based solution and always it is also not easy to find the value of the integral.

SOLVED EXAMPLES

using Picard's me

Example 10.7. Solve $\frac{dy}{dx} = x + y^2$, it is given that $y = 1$, when $x = 0$.

Solution: We know $y_1 = y_0 + \int_{x_0}^{x} f(x,y)\,dx$

$$y_1 = y_0 + \int_{x_0}^{x} (x + y^2)\, dx$$

Here, $y_0 = 1$ and $x_0 = 0$, so we get $f(x, y_0) = x + 1$

$$y_1 = 1 + \int_{x_0=0}^{x} (x + 1)\, dx$$

$$= 1 + \left[\frac{x^2}{2} + x\right]_0^x = 1 + \frac{x^2}{2} + x$$

Now $$y_2 = y_0 + \int_{x_0}^{x} f(x + y_1)\, dx$$

$$= 1 + \int_0^x x + \left(1 + \frac{x^2}{2} + x\right)^2 dx$$

$$y_2 = 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5$$

Example 10.8. Find the value of y, when $x = 0.25, 0.5, 1.0$ and $\frac{dy}{dx} = \frac{x^2}{y^2+1}$ with the initial condition $y = 0$ when $x = 0$.

Solution: $$f(x, y) = \frac{x^2}{y^2+1}; \quad x_0 = 0, y_0 = 0 \text{ so } f(x, y_0) = x^2$$

$$y_1 = y_0 + \int_{x_0}^{x} (x, y_0)\, dx = 0 + \int_0^x x^2\, dx = \frac{x^3}{3}$$

and $$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1)\, dx = 0 + \int_0^x \frac{x^2}{\left(\frac{x^3}{3}\right)^2 + 1}\, dx$$

$$= \tan^{-1}\left(\frac{x^3}{3}\right)$$

$$y_2 = \frac{1}{3}x^3 - \frac{1}{81}x^9 + \ldots.$$

Now $$y(0.25) = \frac{1}{3}(0.25)^3 - \frac{1}{81}(0.25)^9$$

$$= .00520833$$

$$y(0.5) = \frac{1}{3}(.05)^3 - \frac{1}{81}(0.5)^9 = 0.0416666$$

$$y(1.0) = \frac{1}{3}(1.0)^3 - \frac{1}{81}(1.0)^9 = 0.321$$

Example 10.9. **Solve $\frac{dy}{dx} = 2x + z$, $\frac{dz}{dx} = 3xy + x^2z$ given $y = 2$, $z = 0$ when $x = 0$.**

Solution: $y_0 = 2$, $z_0 = 0$ and $x_0 = 0$. So

$$y_1 = y_0 + \int_{x_0}^{x} (2x + z_0)\, dx = 2 + \int_0^x 2x\, dx = 2 + x^2$$

$$z_1 = z_0 + \int_{x_0}^{x} (3xy_0 + x^2z_0)\, dx = 0 + \int_0^x (3x \times 2 + x^2 \times 0)\, dx = 3x^2$$

$$y_2 = y_0 + \int_0^x (2x + z_1)\, dx = 2 + \int_0^x (2x + 3x^2)\, dx = 2 + x^2 + x^3$$

$$z_2 = z_0 + \int_0^x (3xy_1 + x^2z_1)\, dx = \int_0^x [3x \,.\, (2 + x^2) + x^2 \,.\, 3x^2]\, dx$$

$$= 3x^2 + \frac{3}{4}x^4 + \frac{3}{5}x^5$$

Similarly, we get

$$y_3 = 2 + x^2 + x^3 + \frac{3x^5}{20} + \frac{x^6}{10}$$

$$z_3 = 3x^2 + \frac{3x^4}{4} + \frac{3}{5}x^5 + \frac{3}{28}x^4 + \frac{3}{40}x^8.$$

Example 10.10. **Use Picard's method to approximate the value of y when $x = 0.1$, 0.3, 0.4 and 0.5 given that $y = 1$ at $x = 0$ and $y = 1 + xy$, correct to three decimal pla**

Solution: Given $\frac{dy}{dx} = 1 + xy\,;\, y(0) = 1$

$\therefore$ $f(x, y) = 1 + xy,\ y_0 = 1,\ x_0 = 0$

First approximation:

$$y_1 = y_0 + \int_0^x f(x, y_0)\, dx = 1 + \int_0^x (1 + xy_0)\, dx$$

$$= 1 + \int_0^x (1 + x)\, dx = 1 + x + \frac{x^2}{2}$$

Second approximation:

$$y_2 = y_0 + \int_0^x f(x, y_1)\, dx = 1 + \int_0^x (1 + xy_1)\, dx$$

EXERCISE 10.1

Solve the following differential equation by Taylor's series method.

1. $y' = x^2 + y^2$, $y(1) = 0$, find $y(1.3)$.

2. $y' = 1 - \frac{y}{x}$, $y(2) = 2$, find $y(2.1)$. [Ans. $y(1.3) = 0.4158$]

3. $y' = xy^{1/3}$, $y(1) = 1$, find $y(1.1)$, $y(1.2)$. [Ans. $y(2.1) = 0.232$]

4. $y' = xy - 1$, $y(1) = 2$, find $y(1.02)$. [Ans. $y(1.1) = 1.10682$]

5. $y' = y \sin x + \cos x$, $y(0) = 0$. [Ans. $y(1.02) = 2.02061$]

   [Ans. $y = x + \frac{x^3}{6} + \frac{x^5}{120}$]

6. Find $y(0.1)$ by Taylor's series method given that $y' = xy + 1$, $y(0) = 1$.

   [Ans. $\left(y(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} + \frac{x^5}{15}, y(0.1) = 1.1053\right)$]

7. If $y' = \frac{1}{x^2 + y^2}$ with $y(4) = 4$, find $y(4.1)$ and $y(4.2)$ by Taylor's series method.

   [Ans. $y(1.4) = 4.005$, $y(4.2) = 4.0098$.]

8. If $y' = x + y$ and $y(1) = 0$, find $y(1.2)$ by Taylor's series method.

   [Ans. $y(1.2) = 0.232$]

9. Using Picard's method find $y(0.2)$ given that $y' = x - y$; $y(0) = 1$. [Ans. $y(0.2) = 0.837$]

10. Using Picard's method obtain a solution upto the fifth approximation to the equation $y' = y + x$, such that $y(0) = 1$. Check your answer by finding the exact particular solution. Also $y(0.1)$ and $y(0.2)$.

    [Ans. $y(0.1) = 1.1103$; $y(0.2) = 1.2428$]

11. Using Picard's method find $y(0.2)$ and $y(0.4)$ given that $y' = 1 + y^2$ and $y(0) = 0$.

    [Ans. $y(0.2) = 0.2027$, $y(0.4) = 0.4227$]

12. Use Picard's method to approximate the value of y when $x = 0.1$ given that $y(0) = 1$ and $y' = 3x + y^2$.

    [Ans. $y(0.1) = 1.127$]

13. Using Picard's method find the approximate values of y and z corresponding to $x = 0.1$ given that $y(0) = 2$, $z(0) = 1$ and $\frac{dy}{dx} = x + z$, $\frac{dz}{dx} = x - y^2$. [Ans. $y(0.1) = 2.0845$; $z(0.1) = 0.5867$]

14. Using Picard's method obtain the second approximation to the solution to $y'' - x^3y' - x^3y = 0$ so that $y(0) = 1$, $y'(0) = 0.5$.

    [Ans. $y_2 = 1 + \frac{1}{2}x + \frac{3}{40}x^5$.]

10.4 EULER'S METHOD

Consider the equation $\frac{dy}{dx} = f(x, y)$ ...(i)

given that $y(x_0) = y_0$

Its curve of solution through $P(x_0, y_0)$ is shown dotted in Figure 10.1. Now we have to find the ordinate of any other point Q on this curve.

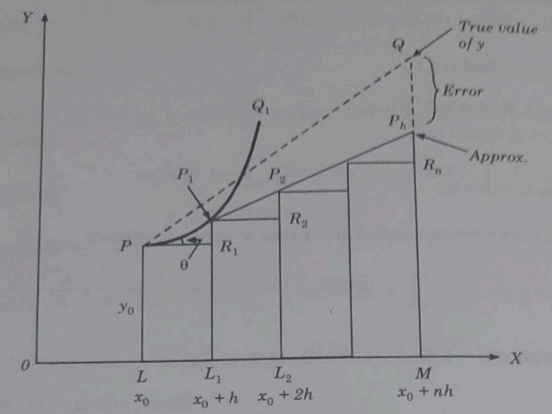

Fig. 10.1

Let us divide LM into n sub-intervals each of width h at $L_1, L_2, \ldots$, so that h is quite small.

In the interval LL_1, we approximate the curve by the tangent at P. If the ordinate through L_1 meet to this tangent in $P_1(x_0 + h, y_1)$, then

$$y_1 = L_1P_1 = LP = R_1P_1 = y_0 + PR_1 \tan q = y_0 + h\left(\frac{dy}{dx}\right)_P = y_0 + h\,(x_0, y_0)$$

Let P_1Q_1 be the curve to solution of (*i*) through P_1 and let its tangent at P_1 meet the ordinate through L_2 in $P_2(x_0 + 2h, y_2)$. Then

$$y_2 = y_1 + h\, f(x_0 + h, y_1) \quad \ldots (ii)$$

Repeating this process n times, we finally reach an approximately MP_n of MQ given by

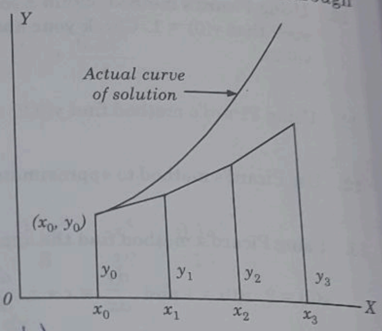

Fig. 10.2.

$$y_n = y_{n-1} + hf(x_0 + \overline{n-1}\,h, y_{n-1})$$

This is Euler's method of finding an approximate solution to (*i*).

10.5 DRAWBACKS OF EULER'S METHOD

A great disadvantage of the above method lies in the fact that if $\frac{dy}{dx}$ changes rapidly over an interval, then its value at the beginning of the interval may give a poor approximation as

$$y_3^{(4)} = 1.52535 + \frac{0.2}{2}\left[0.4 + 1\left|\sqrt{1.52535}\right| + \left(0.6 + \left|\sqrt{1.88619}\right|\right)\right]$$

$$= 1.88619, \text{ correct to 5 decimal places.}$$

Since, $y_3^{(3)} = y_3^{(4)} = 1.88619$. Hence, we take $y = 1.88619$ at $x = 0.6$.

10.6 MODIFIED EULER'S METHOD

In the Euler's method the curve of solution in the interval LL_1 is approximated by the tangent at P(figure earlier) such that at P_1, we have

$$y_1 = y_0 = hf(x_0, y_0) \qquad \ldots(i)$$

Then the slope of the curve of solution through P_1 [*i.e.* $(dy/dx)\ P_1 = f(x_0 + h, y_1)$ is computed and the tangent at P_1 to P_1Q_1 is drawn meeting the ordinate through L_2 in $P_2(x_0 + 2h, y_2)$.

Now we find a better approximation $y_1^{(1)}$ of $y(x_0 + h)$ by taking the slope of the curve as the mean of the slopes of the tangents at P and P_1, *i.e.*,

$$y_1^{(1)} = y_0 + \frac{h}{2}\,[f(x_0, y_0) + f(x_0, h, y_1)] \qquad \ldots(ii)$$

As the slope of the tangent at P_1 is not known, we take y_1 as found in (*i*) by Euler's method it on R.H.S. of (*ii*) to obtain the first modified value $y_1(1)$. The equation (*i*) is therefore, predictor while (*ii*) serves as the corrector of y_1.

Again the corrector is applied and we find a still better value $y_1^{(2)}$ corresponding to L_1 as

$$y_1^{(2)} = y_0 + \frac{h}{2}\,[f(x_0, y_0) + f(x_0, h, y_1^{(1)})]$$

We repeat the step, till two consecutive values of y agree. This is then taken as the starting point for the next interval L_1L_2.

Once y_1 is obtained to desired degree of accuracy, y corresponding to L_2 is found from the predictor.

$$y_2 = y_1 + hf(x_0 + h, y_1)$$

and a better approximation $y_2^{(1)}$ is obtained from the corrector

$$y_2^{(1)} = y_1 + \frac{h}{2}\,[f(x_0 + h, y_1) + f(x_0 + 2h, y_2)]$$

We repeat this step until y_2 becomes stationary. Then we proceed to calculate y_3 as above and so on.

This is the *modified Euler's method* which is a predictor-corrector method.

SOLVED EXAMPLES

Example 10.18. Solve $\dfrac{dy}{dx} = 1 - y$, $y(0) = 0$ in the range $0 \le x \le 0.2$ by taking $h = 0.1$.

Solution: By Euler's method

$$y_1^{(1)} = y_0 + hf(x_0, y_0) = 0 + (0.1)(1 - 0) = 0.1$$

Now by formula,

$$y_1 = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1), y_1^{(1)}]$$

$$= 0 + \frac{(0.1)}{2}[1 + (1 - 0.1)] = 0.095$$

This y_1 is the value of y corresponding to $x = x_1$ by modified Euler's method. Now we use the formula iteratively to get even better approximations to y_1. Thus if 0.095 is the first approximation to y_1 then the second approximation to y_1

$$= y_0 + \left(\frac{h}{2}\right)[f(x_0, y_0) + f(x_1, 0.095)]$$

$$= 0 + \left(\frac{1}{2}\right)[1 + 1 - 0.095] = 0.0952$$

Third approximation to y_1

$$= 0 + \left(\frac{h}{2}\right)[f(x_0, y_0) + f(x_1, 0.0952)]$$

$$= \left(\frac{1}{2}\right)[1 + 1 - 0.0952] = 0.09524.$$

Now we take $y_1 = 0.9524$ and proceed on to find $y_2 = y(x = 0.2)$. The value of y_2 by Euler's method is given by

$$y_2^{(1)} = y_1 + hf(x_1, y_1) = 0.09524 + (0.1)[1 - 0.09524]$$

$$= 0.185716.$$

Hence by (8.8) the first approximation to y_2 is given by

$$= y_1 + \left(\frac{h}{2}\right)[f(x_1, y_1) + f(x_2, y_2), y_2^{(1)}]$$

$$= 0.09524 + \left(\frac{0.1}{2}\right)[(1 - 0.09524) + (1 - 0.185716)]$$

$$= 0.09524 + 0.0859522 = 0.1811922.$$

The second approximation to y_2

$$= y_1 + \left(\frac{h}{2}\right)[f(x_1, y_1) + f(x_1, 0.1811922)]$$

$$= 0.09524 + \left(\frac{0.1}{2}\right)[(1 - 0.09524) + (1 - 0.1811922)]$$

$$= 0.18141839.$$

Example 10.19 **Using Euler's modified, obtain a solution to the equation $\frac{dy}{dx} = x + \left|\sqrt{y}\right| = f(x, y)$ with initial condition $y = 1$ at $x = 0$ for the range $0 \le x \le 0.6$ in steps of 0.2.**

Solution: Here $x_0 = 0, y_0 = 1, h = 0.2$

By Euler's method, we have

$$y_1^{(1)} = y_0 + hf(x_0, y_0) = 1 + 0.2[1 + 0] = 1.2.$$

First corrected value of y_1

$$= 1 + \frac{1}{2}(0.2) + [(1 + 0) + \{0.2 + \sqrt{1.2}\}]$$
$$= 1 + 0.1\,[1.2 + 1.095] = 1.2295$$

Second corrected value of y_1 $= 1 + \frac{1}{2}(0.2) + [(1 + 0) + \{0.2 + \sqrt{(1.2295)}\}] = 1.2309$

Third corrected value of $y_1 = 1 + \frac{1}{2}(0.2) + [(1 + 0) + \{0.2 + \sqrt{(1.2309)}\}] = 1.2309$

Hence we accept $y_1 = 1.2309$.

Now again by Euler's method,

$$y_2^{(1)} = y_1 + hf(x_1, y_1) = 1.2309 + 0.2[0.2 + \sqrt{(1.2309)}]$$
$$= 1.2309 + 0.2\,[0.2 + 1.109] = 1.4927$$

First corrected value of y_2 $= 1.2309 + \frac{1}{2}(0.2)\,[\{0.2 + \sqrt{(1.2309)}] + \{0.4 + \sqrt{(1.4927)}\}]$

$$= 1.2309 + (0.1)\,[0.6 + 1.109 + 1.222] = 1.5240$$

Second corrected value of y_2 $= 1.2309 + (0.1)\,[\{(0.2 + \sqrt{(1.2309)} + \{0.4 + \sqrt{(1.5240)}]]$

$$= 1.2309 + (0.1)\,[0.6 + 1.109 + 1.235] = 1.5253.$$

Third corrected value of y_2 $= 1.2309 + (0.1)[0.6 + \sqrt{(1.2309)} + \sqrt{(1.5253)}] = 1.5253$

Hence we accept $y_2 = 1.5253$

Now again by Euler's method, the value of y for $x = 0.6$ is given by

$$y_3^{(1)} = y_2 + (0.2)\,(x_2 + \sqrt{y_2})$$
$$= 1.5253 + (0.2)\,[(0.4) + \sqrt{(1.5253)}]$$
$$= 1.5253 + (0.2)\,[0.4 + 1.235] = 1.8523$$

Hence, first corrected value of y_3

$$= 1.5253 + \frac{1}{2}(0.2)\,[\{(0.4) + \sqrt{(1.5253)}\} + \{0.6 + \sqrt{(1.8523)}\}]$$
$$= 1.5253 + (0.1)\,[1 + 1.235 + 1.361] = 1.8849.$$

Second corrected value of y_3

$$= 1.5253 + (0.1)\,[1 + \sqrt{(1.5253)}] + \sqrt{(1.8849)}]$$
$$= 1.5253 + (0.1)\,\{1 + 1.235 + 1.373\} = 1.8851$$

Third corrected value will also be 1.8851

Hence for $x = 0.6$, we have $y = y_3 = 1.8851$.

Example 10.20. **Find $y(2.2)$ using Euler's modified method for**

$$\frac{dy}{dx} = -xy^2, \text{ where } y(2) = 1 \text{ (Take } h = 0.1)$$

Solution: By Euler's method, we obtain

$$y_1^{(0)} = y_1 = y_0 + hf(x_0, y_0) = 1 + 0.1(-2)(-1)^2 = 0.8$$

7. Solve the differential equation: $\frac{dy}{dx} = -xy^2$, $y = 2$ at $x = 0$, by modified Euler's method and obtain y at $x = 0.2$ in two stages of 0.1 each. [Ans. 1.9227]

8. Given that $\frac{dy}{dx} = 2 + \sqrt{(xy)}$ and $y = 1$ when $x = 1$, find approximate value of y at $x = 2$ in steps of 0.2, using Euler's modified method. [Ans. 5.051]

10.7 RUNGE-KUTTA METHOD

Taylor's series method of solving differential equation numerically is restricted by the labour involved in finding the higher order derivatives. However, there is a class of methods known as Runge-Kutta methods which do not require the calculations of higher order derivatives and give greater accuracy. The Runge-Kutta formulae possess the advantages of requiring only the function values at some selected points. This class methods agrees with Taylor's series solution upto the term h^r where r differs from method to method and is called the order of that method.

(*i*) First order Runge-Kutta Method: Consider the first order differential equation.

$$\frac{dy}{dx} = f(x, y),\ y(x_0) = y_0 \quad \text{... (i)}$$

By Euler's method we know that

$$y_1 = y_0 + hf(x_0, y_0)$$

Expanding by Taylor's series, we get

$$y_1 = y(x_0 + h) = y_0 + h\,y_0' + \frac{h^2}{2!}\,y_0'' + \quad \text{... (ii)}$$

It follows that the Euler's method agrees with the Taylor's series solution upto the term in h, hence, Euler's method is the Runge-Kutta method of first order.

(*ii*) Second Order Runge-Kutta Method: We know that modified Euler's method.

$$y_1 = y_0 + \frac{h}{2}\,[f(x_0, y_0) + f(x_0, h, y_1)] \quad \text{... (iii)}$$

Putting $y_1 = y_0 + hf(x_0, y_0)$ on the right hand side of equation (*iii*), we get,

$$y_1 = y_0 + \frac{h}{2}\,[f_0 + f(x_0 + h, y_0 + hf_0)] \quad \text{... (iv)}$$

[where $f_0 = f(x_0, y_0)$]

Expanding left hand side by Taylor's series, we get

$$y_1 = y(x_0 + h) = y_0 + h\,y_0' + \frac{h^2}{2!}\,y_0'' + \frac{h^3}{3!}\,y_0''' + \ldots \quad \text{... (v)}$$

Expanding $f(x_0 + h, x_0 + hf_0)$ by Taylor's series for a function of two variables, equation (*iv*) gives

$$y_1 = y_0 + \frac{h}{2}\left[f_0 + \left\{f(x_0, y_0) + h\left(\frac{\partial f}{\partial x}\right)_0 + hf_0\left(\frac{\partial f}{\partial y}\right)_0 + 0(h^2) + \ldots\right\}\right]$$

$$= y_0 + \frac{1}{2}\left[hf_0 + hf_0 + h^2\left\{\left(\frac{\partial f}{\partial x}\right)_0 + \left(\frac{\partial f}{\partial y}\right)_0\right\} + 0(h^3)\right]$$

7. Solve the differential equation: $\frac{dy}{dx} = -xy^2$, $y = 2$ at $x = 0$, by modified Euler's method and obtain y at $x = 0.2$ in two stages of 0.1 each. [Ans. 1.9227]

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Taylor's series method of solving differential equation numerically is restricted by the labour involved in finding the higher order derivatives. However, there is a class of methods known as Runge-Kutta methods which do not require the calculations of higher order derivatives and give greater accuracy. The Runge-Kutta formulae possess the advantages of requiring only the function values at some selected points. This class methods agrees with Taylor's series solution upto the term h^r where r differs from method to method and is called the order of that method.

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It follows that the Euler's method agrees with the Taylor's series solution upto the term in h, hence, Euler's method is the Runge-Kutta method of first order.

(ii) Second Order Runge-Kutta Method: We know that modified Euler's method.

$$y_1 = y_0 + \frac{h}{2}\,[f(x_0, y_0) + f(x_0, h, y_1)] \quad \text{... (iii)}$$

Putting $y_1 = y_0 + hf(x_0, y_0)$ on the right hand side of equation (*iii*), we get,

$$y_1 = y_0 + \frac{h}{2}\,[f_0 + f(x_0 + h, y_0 + hf_0)] \quad \text{... (iv)}$$

[where $f_0 = f(x_0, y_0)$]

Expanding left hand side by Taylor's series, we get

$$y_1 = y(x_0 + h) = y_0 + h\,y_0' + \frac{h^2}{2!}\,y_0'' + \frac{h^3}{3!}\,y_0''' + \ldots \quad \text{... (v)}$$

Expanding $f(x_0 + h, x_0 + hf_0)$ by Taylor's series for a function of two variables, equation (*iv*) gives

$$y_1 = y_0 + \frac{h}{2}\left[f_0 + \left\{f(x_0, y_0) + h\left(\frac{\partial f}{\partial x}\right)_0 + hf_0\left(\frac{\partial f}{\partial y}\right)_0 + 0(h^2) + \ldots\right\}\right]$$

$$= y_0 + \frac{1}{2}\left[hf_0 + hf_0 + h^2\left\{\left(\frac{\partial f}{\partial x}\right)_0 + \left(\frac{\partial f}{\partial y}\right)_0\right\} + 0(h^3)\right]$$

$$= y_0 + hf_0 + \frac{h^2}{2} f_0' + 0(h^3) \qquad \left[\because \frac{df(x,y)}{dx} = \frac{\partial f}{\partial x} = f\frac{\partial f}{\partial y}\right]$$

$$= y_0 + h\, y_0' + \frac{h^2}{2!}\, y_0'' + 0(h^3) \qquad \text{... }(vi)$$

Equating (*v*) and (*vi*), it follows that the modified Euler's method agrees with Taylor's series solution upto the term in h^2.

Hence, the modified Euler's method is the Runge-Kutta method of second order.

Therefore, the second order Runge-Kutta formula is

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2), \text{ where } k_1 = hf(x_0, y_0)$$

and

$$k_2 = hf(x_0 + h, y_0 + k_1)$$

***(iii)* Third Order Runge-Kutta Method:** On the lines of above methods, one can see that Runge-Kutta method agrees with the Taylor's series solution upto the terms in h^3, *i.e.*,

$$y_1 = y_0 + \frac{h}{6}\left[f(x_0,y_0) + 4f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} f(x_0,y_0)\right) + f(x_0 + h, y_0 + hf(x_0 + h, y_0 + hf(x_0,y_0))\right] \qquad \text{...}(vii)$$

The third order Runge-Kutta formula is

$$y_1 = y_0 + \frac{1}{6}(k_1 + 4k_2 + k_3)$$

where

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

Now, from differential equation $y' = \frac{dy}{dx} = f(x, y)$, we obtain

$$y'' = f_x + f_y y' = f_x + ff_y = F_1$$

$$y''' = f_{xx} + 2ff_{xy} + f^2 f_{yy} + f_x f_y + ff_y^2$$

Similarly,

Putting these values in (*viii*), we get

$$y(x+h) = y_n + nf + \frac{h^2}{2!} \cdot F_1 +$$

Using the above notation and Taylor's theorem,

$$k_1 = hf$$

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2), \text{ where } k_1 = hf(x_0, y_0)$$

and $$k_2 = hf(x_0 + h, y_0 + k_1)$$

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$$y_1 = y_0 + \frac{h}{6}\left[f(x_0, y_0) + 4f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}f(x_0, y_0)\right) + f(x_0 + h, y_0 + hf(x_0 + h, y_0 + hf(x_0, y_0))\right] \quad ...(vii)$$

The third order Runge-Kutta formula is

$$y_1 = y_0 + \frac{1}{6}(k_1 + 4k_2 + k_3)$$

where

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = hf(x_0, h, y_0 + k')$$

where

$$k' = hf(x_0, h, y_0 + k_1)$$

***(iv)* Fourth Order Runge-Kutta Method:** The Runge-Kutta fourth order method is considered with the Taylor's Series solution upto terms of h^4. We know that Taylor's series solution can be expressed in terms of $f(x, y)$ and its partial derivatives of various orders. Here we take

$$\begin{aligned} k_1 &= hf(x, y) \\ k_2 &= hf(x, mh, y + mk_1) \\ k_3 &= hf(x, nh, y + nk_2) \\ k_4 &= hf(x, ph, y + pk_3) \end{aligned} \quad ...(viii)$$

to avoid the calculations of derivatives

Now, $$y(x + h) = y(x) + ak_1 + bk_1 + ck_3 + dk_4 \quad ...(ix)$$

We know that by Taylor series

$$y(x + h) = y(x) + hy' + \frac{h^2}{2!}y'' + \frac{h^3}{3!}y''' + \frac{h^4}{4!}y^{iv} + O(h^5) \quad ...(x)$$

we put it in a convenient form, *i.e.*,

$$F_1 = f_x + ff_y,\ F_2 = f_{xx} + 2ff_{xy} + f^2 f_{yy}$$

$$F_3 = f_{xxx} + 3ff_{xxy} + 3f^2 = f_{xyy} + f^3 f_{yyy}$$

Using the above notation and Taylor's theorem, similarly we get ...(xi)

$$k_1 = hf$$

$$k_2 = h\left[f + mhF_1 + \frac{1}{2}m^2h^2F_2 + \frac{1}{6}m^3h^3F_3 + ...\right] \quad [\text{where } f = f(x, y)]$$

$$k_3 = h\left[f + mhF_1 + \frac{1}{2}h^2(n^2F_2 + 2mn + f_yF_1 + \frac{1}{6}h^3(n^3F_3 + 3m^2n \cdot f_yF_2 + 6mn^2(f_{xy} + ff_{yy})F_1) + ...\right]$$

and

$$k_4 = \left[f + phF_1 + \frac{1}{2}h^2(p^2F_2 + 2npf_yF_1) + \frac{1}{6}h^3(p^3F_3 + 3n^2pf_yF_2 + 6np^2(f_{xy} + ff_{yy})F_1 + 6mnpf_y^2F_1) + ...\right]$$

Substituting these values in (ix), we get

$$y(x_0 + h) = y(x) + (a + b + c + d)hf + (bm + cn + dp)h^2F_1 + \frac{1}{2}(bm^2 + cn^2 + dp^2)h^3f_yF_1 + \frac{1}{2}(cm^2n + dn^2p)h^4f_yF + (cm\,n^2 + dnp^2)h_4(f_{x} + ff_{yy})F_1 + dmnph^4f_y^2F_1 + O(h^5)$$

Substituting with (xi), we get

$$a + b + c + d = 1, \qquad cmn + dnp = \frac{1}{6}$$

$$bm + cn + dp = \frac{1}{2}, \qquad cmn^2 + dnp^2 = \frac{1}{8}$$

$$bm^2 + cn^2 + dp^2 = \frac{1}{3}, \qquad cm^2n + dn^2p = \frac{1}{12}$$

$$bm^3 + cn^3 + dp^3 = \frac{1}{4}, \qquad dmnp = \frac{1}{24}$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 4k_2 + k_3)$$

where $\quad k_1 = hf(x_0, y_0)$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = hf(x_0, h, y_0 + k')$$

where $\quad k' = hf(x_0, h, y_0 + k_1)$

***(iv)* Fourth Order Runge-Kutta Method:** The Runge-Kutta fourth order method is considered with the Taylor's Series solution upto terms of h^4. We know that Taylor's series solution can be expressed in terms of $f(x, y)$ and its partial derivatives of various orders. Here we take

$$\begin{aligned} k_1 &= hf(x, y) \\ k_2 &= hf(x, mh, y + mk_1) \\ k_3 &= hf(x, nh, y + nk_2) \\ k_4 &= hf(x, ph, y + pk_3) \end{aligned} \qquad \text{... (viii)}$$

to avoid the calculations of derivatives

Now, $\quad y(x + h) \approx y(x) + ak_1 + bk_1 + ck_3 + dk_4 \qquad$ ... (ix)

We know that by Taylor series

$$y(x + h) = y(x) + hy' + \frac{h^2}{2!}y'' + \frac{h^3}{3!}y''' + \frac{h^4}{4!}y^{iv} + 0(h^5) \qquad \text{... (x)}$$

we put it in a convenient form, *i.e.*,

$$F_1 = f_x + ff_y,\ F_2 = f_{xx} + 2ff_{xy} + f^2 f_{yy}$$

$$F_3 = f_{xxx} + 3\,ff_{xxy} + 3f^2 = f_{xyy} + f^3 f_{yyy}$$

$$+\frac{1}{6}h^3(h^3F_3 + 3m^2n\,.f_yF_2 + 6mn^2(f_{xy} + ff_{yy})F_1] + \ldots\Big]$$

and

$$k_4 = \Big[f + phF_1 + \frac{1}{2}h^2(p^2F_2 + 2npf_yF_1)$$

$$+\frac{1}{6}h^3(p^3F_3 + 3n^2pf_yF_2 + 6np^2(f_{xy} + ff_{yy})F_1 + 6mnpf_y^2F_1$$

Substituting these values in (*ix*), we get

$$y(x_0 + h) = y(x) + (a + b + c + d)\,hf + (bm + cn + dp)h^2F_1$$

$$+\frac{1}{2}(bm^2 + cn^2 + dp^2)h^3f_yF_1 + \frac{1}{2}(cm^2n +$$

$$+ (cm\,n^2 + dnp^2)h_4(f_n + ff_{yy}))F_1 + dmm\,ph^4f_y^2F_1$$

Substituting with (*xi*), we get

$$a + b + c + d = 1, \qquad cmn + dnp = \frac{1}{6}$$

$$bm + cn + dp = \frac{1}{2}, \qquad cmn^2 + dnp^2 = \frac{1}{8}$$

$$bm^2 + cn^2 + dp^2 = \frac{1}{3}, \qquad cm^2n + dn^2p = \frac{1}{12}$$

$$bm^3 + cn^3 + dp^3 = \frac{1}{4}, \qquad dmnp = \frac{1}{24}$$

Now, from differential equation $y' = \frac{dy}{dx} = f(x, y)$, we obtain

$$y'' = f_x + f_y y' = f_x + ff_y = F_1$$

$$y''' = f_{xx} + 2ff_{xy} + f^2 f_{yy} + f_x f_y + ff_y^2$$

$$= (f_{xx} + 2ff_{xy} + f^2 f_{yy}) + f_y (f_x + ff_y) = F_2 + f_y F_1$$

Similarly,

$$y^{iv} = F_3 + f_y F_2 + 3F_1 (f_{xy} + ff_{xy}) + f_y^2 F_1$$

Putting these values in (*viii*), we get

$$y(x + h) = y_n + nf + \frac{h^2}{2!} \cdot F_1 + \frac{1}{6} h^3 (F_2 + f_y F_1) + \frac{1}{24} h^4 [F_3 + f_y F_2$$

$$+ 3(f_{xy} + ff_{yy}) F_1 + f_y^2 \cdot F_1] + \ldots \quad \ldots (xi)$$

Using the above notation and Taylor's theorem, similarly we get [where $f = f(x, y)$]

$$k_1 = hf$$

$$k_2 = h\left[f + mhF_1 + \frac{1}{2}m^2h^2F_2 + \frac{1}{6}m^3h^3F_3 + \ldots\right]$$

$$k_3 = h\left[f + mhF_1 + \frac{1}{2}h^2(n^2F_2 + 2mn + f_yF_1\right.$$

$$\left.+\frac{1}{6}h^3(n^3F_3 + 3m^2n \,.\, f_yF_2 + 6mn^2[f_{xy} + ff_{yy})F_1] + \ldots\right]$$

and

$$k_4 = \left[f + phF_1 + \frac{1}{2}h^2(p^2F_2 + 2npf_yF_1)\right.$$

$$\left.+\frac{1}{6}h^3(p^3F_3 + 3n^2pf_yF_2 + 6np^2(f_{xy} + ff_{yy})F_1 + 6mnpf_y^2F_1) + \ldots\right]$$

Substituting these values in (*ix*), we get

$$y(x_0 + h) = y(x) + (a + b + c + d)\, hf + (bm + cn + dp)h^2 F_1$$

$$+\frac{1}{2}(bm^2 + cn^2 + dp^2).h^3 f_y F_1 + \frac{1}{2}(cm^2n + dn^2p)h^4 f_y F_2$$

$$+ (cm\, n^2 + dnp^2)h_4 (f_n + ff_{yy}))\, F_1 + dmm\, ph^4 f_y^2 F_1 + O(h^5) \quad \ldots (xii)$$

Substituting with (*xi*), we get

$$a + b + c + d = 1, \qquad cmn + dnp = \frac{1}{6}$$

$$bm + cn + dp = \frac{1}{2}, \qquad cmn^2 + dnp^2 = \frac{1}{8}$$

$$bm^2 + cn^2 + dp^2 = \frac{1}{3}, \qquad cm^2n + dn^2 p = \frac{1}{12}$$

$$bm^3 + cn^3 + dp^3 = \frac{1}{4}, \qquad dmnp = \frac{1}{24}$$

These are eight equations in seven unknowns; and classical solution to these equation is

$$m = n = \frac{1}{2}, p = 1, a = d = \frac{1}{6}, b = c = \frac{1}{3}.$$

Putting these values in (*viii*) and (*ix*), the Runge-Kutta formula reduces to

$$\left.\begin{aligned} k_1 &= hf(x, y) \\ k_2 &= f\left(x + \frac{h}{2}, y + \frac{h_1}{2}\right) \\ k_3 &= hf\left(x + \frac{h}{2}, y + \frac{h_2}{2}\right) \\ k_4 &= hf(x + h, y + k_3) \end{aligned}\right\} \quad \text{... }(xiii)$$

and
$$y(x + h) = y(x) + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

From this, we have

$$y_1 = y(x_0 + h) = y_0 + \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4]$$

$$\left.\begin{aligned} k_1 &= hf(x_0, y_0) \\ k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \\ k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \\ k_4 &= hf(x_0 + h, y_0 + k_3) \end{aligned}\right\} \quad \text{... }(xiv)$$

SOLVED EXAMPLES

mple 10.24. Use Runge's method to find an approximate value of y when $x = 0.2$ en that

$$\frac{dy}{dx} = x + y = x + y \text{ and } y = 1 \text{ when } x = 0.$$

tion: Here we have

$$x_0 = 0, \quad y_0 = 1, h = 0.2, f(x_0, y_0) = 1$$
$$k_1 = hf(x_0, y_0) = (0.2)(1) = 0.2$$
$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0.2) f(0.1, 1.1) = 0.240$$
$$k' = hf(x_0, h, y_0 + k) = (0.2 f(0.2, 1.2) = 0.280$$

nd
$$k_3 = hf(x_0, h, + y_0 + k') = (0.2 f(0.1, 1.28) = 0.296$$

$$k = \frac{1}{6}[k_1 + 4k_2 + k_3] = \frac{1}{6}[0.200 + 0.960 + 0.296] = 2.2426.$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$
$$k_4 = hf(x_0 + h, y_0 + k_3) \qquad \ldots (xiv)$$

SOLVED EXAMPLES

Example 10.24. **Use Runge's method to find an approximate value of y when $x = 0.2$ given that**

$$\frac{dy}{dx} = x + y = x + y \text{ and } y = 1 \text{ when } x = 0.$$

Solution: Here we have

$$x_0 = 0, \quad y_0 = 1, h = 0.2, f(x_0, y_0) = 1$$

$\therefore$
$$k_1 = hf(x_0, y_0) = (0.2)(1) = 0.2$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0.2) f(0.1, 1.1) = 0.240$$

$$k' = hf(x_0, h, y_0 + k) = (0.2 f(0.2, 1.2) = 0.280$$

and
$$k_3 = hf(x_0, h, + y_0 + k') = (0.2 f(0.1, 1.28) = 0.296$$

$\therefore$
$$k = \frac{1}{6}[k_1 + 4k_2 + k_3] = \frac{1}{6}[0.200 + 0.960 + 0.296] = 2.2426.$$

$\therefore$

Example 10.26. **Solve $10 y' = x^2 + y^2, y(0) = 1$** **order Runge-Kutta algorithm.**

Solution: The given equation can be written as

$$y' = \frac{dy}{dx} = \frac{x^2 + }{1}$$

We have
$$f(x, y) = \frac{x^2 + y^2}{10}$$

To find $y(0.2)$

Here
$$x_0 = 0, \quad y_0$$

$$k_1 = hf(x_0, y_0) = 0.2$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) =$$

Hence the required approximate value of

$$y = y_0 + k = 1 + 0.2426 = 1.2426.$$

Example 10.25. **Find an approximate value of y when $x = 0.8$ for the particular solution of $\frac{dy}{dx} = \sqrt{x+y}$ satisfying $y = 0.41$ when $x = 0.40$, using Runge-Kutta method.**

Solution: Here

$$h = 0.8 - 0.4 = 0.4$$

$$x_0 = 0.40,\ y_0 = 0.41$$

$$f(x, y) = \sqrt{x+y}$$

$$k_1 = hf(x_0, y_0) = 0.4\ \sqrt{0.4+0.41} = 0.36$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0.4)\ \sqrt{0.4+0.2+0.41+0.18} = 0.43635$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = (0.4)\ \sqrt{0.6+0.41+0.21817} = 0.44329$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = (0.4)\ \sqrt{0.8+0.41+0.44329} = 0.51432$$

Hence the required value of y by this formula is

$$y = y_0 + k \text{ where } y_0 = 0.41$$

$$k = \frac{1}{6}[k_1 + k_4 + 2(k_2 + k_3)] = \frac{1}{6}[0.36 + 0.51432 + 0.8727 + 0.88638]$$

$$= 0.43693$$

$\therefore$ $y = 0.41 + 0.43693 = 0.84893.$

Example 10.26. **Solve $10\,y' = x^2 + y^2$, $y(0) = 1$ to evaluate $y(0.2)$ and $y(0.4)$ by fourth Runge-Kutta algorithm.**

EXERCISE 10.3

1. Apply Runge-Kutta fourth order method, to find an approximate value of y where $x = 0.2$ given that $\frac{dy}{dx} = x + y$ and $y = 1$ when $x = 0$. **[Ans.** 1.2428]
2. Using Runge-Kutta method of order 4, find (0.2) for the equation $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$. Take $h = 0.2$. **[Ans.** 1.1678]
3. Find by Runge-Kutta method an approximate value of y for $x = 0.8$, given that $y = 0.41$ when $x = 0.4$ and $\frac{dy}{dx} = \sqrt{(x+y)}$. **[Ans.** 0.8489]
4. Using Runge-Kutta method of order 4, find $y(0.2)$ given that $\frac{dy}{dx} = 3x + \frac{1}{2}y$, $y(0) = 1$, taking $h = 0.1$. **[Ans.** 1.1749]
5. Using Runge-Kutta method, solve $y'' = xy'^2 = xy^2 - y^2$ for $x = 0.2$ correct to 4 decimal places. Initial conditions are $x = 0$, $y = 1$, $y' = 0$. **[Ans.** $y^{(0.2)} = 0.9801$, $y'^{(0.2)} = -0.1970$]
6. Solve the differential equations: $\frac{dy}{dx} = 1 + xz$, $\frac{dz}{dx} = -xy$ for $x = 0.3$, by using fourth order Runge-Kutta method. Initial values are $x = 0$, $y = 0$, $z = 1$. **[Ans.** $y = 0.3448$, $z = 0.99$]

10.8 PREDICTOR-CORRECTOR METHODS

Predictor-corrector methods refer to a family of schemes for solving ordinary differential equations using two formulae: *predictor* and *corrector* formulae. In predictor-corrector methods, four prior values are required to find the value of y at x_n. Predictor-corrector methods have the advantage of giving an estimate of error from successive approximations to y_n. The predictor is an explicit formula and is used first to determine an estimate of the solution y_{n+1}. The value y_{n+1} is calculated from the known solution at the previous point (x_n, y_n) using single-step method or several previous points (multiple-step methods). If x_n and x_{n+1} are two consecutive mesh points such that $x_{n+1} = x_n + h$, then in Euler's method we have

$$y_{n+1} = y_n + h\,f(x_0 + nh, y_n),\ n = 0, 1, 2, 3, \ldots \qquad \ldots (i)$$

Once an estimate of y_{n+1} is found, the corrector is applied. The corrector uses the estimated value of y_{n+1} on the right-hand side of an otherwise implicit formula for computing a new, more accurate value of y_{n+1} on the left-hand side.

The modified Euler's method gives as

$$y_{n+1} = y_n + \frac{h}{2}\,[f(x_n, y_n) + f(x_{n+1}, y_{n+1})] \qquad \ldots (ii)$$

The value of y_{n+1} is first estimated by (*i*) and then utilised in the right-hand side of Eq. (*ii*) resulting in a better approximation of y_{n+1}. The value of y_{n+1} thus obtained is again substituted in Eq. (*ii*) to find a still better approximation of y_{n+1}. This procedure is repeated until two consecutive iterative values of y_{n+1} are very close. Here, the corrector equation (*ii*) which is an implicit equation is being used in an *explicit* manner since no solution of a non-linear equation is required.

In addition, the application of corrector can be repeated several times such that the new value of y_{n+1} is substituted back on the right-hand side of the corrector formula to obtain a more refined value for y_{n+1}. The technique of refining an initially crude estimate of y_{n+1} by means of a more accurate formula is known as *predictor-corrector* method. Equation (*i*) is called the *predictor* and Eqn. (*ii*) is called the *corrector* of y_{n+1}. In what follows, we describe two such predictor-corrector methods.

1. Adams-Bashforth method.
2. Milne's method.

10.8.1 Adams-Bashforth Method

The Adams-Bashforth method uses the information at past four solution points to extrapolate the solution at the next point. Therefore, in order to apply Adams-Bashforth method three more solution points, in addition to the starting solution point, are computed using any of the methods discussed in the preceding sections.

In order to derive the mathematical expression for Adams-Bashforth method, we rewrite the first order ordinary differential equation

$$\frac{dy}{dx} = f(x, y) \text{ as } dy = f(x, y)\, dx$$

Integrating the equation between $x = x_k$ and $x = x_{k+1}$, we get

$$\int_{x_k}^{x_{k+1}} dy = \int_{x_k}^{x_{k+1}} f(x, y)\, dx$$

$$\Rightarrow \quad y_{k+1} = y_k + \int_{x_k}^{x_{k+1}} f(x, y)\, dx \qquad \text{... (i)}$$

To solve the integral on the right, we approximate $f(x, y)$ as a polynomial in x. We derive this by making it a fit through the past points (x_{k-3}, y_{k-3}), (x_{k-2}, y_{k-2}), (x_{k-1}, y_{k-1}) and (x_k, y_k). Suppose we approximate it using Newton's backward difference interpolating polynomial as

$$f(x, y) = f_k + u \nabla f_k + \frac{u(u+1)}{2} \nabla^2 f_k + \frac{u(u+1)(u+2)}{6} \nabla^3 f_k \qquad \text{... (ii)}$$

where $u = \dfrac{x - x_k}{h}$, and ∇f_k, $\nabla^2 f_k$, $\nabla^3 f_k$ are backward differences at point $x = x_k$.

Therefore
$$y_{k+1} = y_k + \int_{x_k}^{x_{k+1}} \left[f_k + u\nabla f_k + \frac{u(u+1)}{2} \nabla^2 f_k + \frac{u(u+1)(u+2)}{6} \nabla^3 f_k \right] dx$$

Since $u = \dfrac{x - x_k}{h}$, therefore $dx = h\,du$

And at $x = x_k \quad u = 0$

at $x = x_{k+1} \quad u = 1$

Substituting these values, we get

$$y_{k+1} = y_k + h \int_0^1 \left[f_k + u\nabla f_k + \frac{u(u+1)}{2} \nabla^2 f_k + \frac{u(u+1)(u+2)}{6} \nabla^3 f_k \right] du$$

$$= y_k + h \left[f_k + \frac{1}{2} \nabla f_k + \frac{5}{12} \nabla^2 f_k + \frac{3}{8} \nabla^3 f_k \right] du$$

Expressing the backward differences in terms of function values, and substituting in the above expression, we get

$$y_{k+1}^{p} = y_k + \frac{h}{24}\,[-9f_{k-3} + 37f_{k-2} + 59f_{k-1} + 55f_k] \qquad \dots (iii)$$

This is called *Adams-Bashforth formula*. It is used as a *predictor formula*. The superscript p indicates the predicted value of y_{k+1}.

The corrector formula is developed similarly. Now we construct a Newton's backward difference interpolating polynomial passing through the points (x_{k-2}, y_{k-2}), (x_{k-1}, y_{k-1}), (x_k, y_k) and (x_{k+1}, y^{p}_{k+1}) as

$$f(x, y) = f_{k+1} + u\nabla f_{k+1} + \frac{u(u+1)}{2}\nabla^2 f_{k+1} + \frac{u(u+1)(u+2)}{6}\nabla^3 f_{k+1} \qquad \dots (iv)$$

where $u = \dfrac{x - x_{k+1}}{h}$, and ∇f_{k+1}, $\nabla^2 f_{k+1}$, $\nabla^3 f_{k+1}$ are backward differences at point $x = x_{k+1}$.

Now substituting $f(x, y)$ in (i) and integrating between $x = x_k$ and $x = x_{k+1}$, we get

$$y_{k+1} = y_k + \int_{x_k}^{x_{k+1}} \left[f_{k+1} + u\nabla f_{k+1} + \frac{u(u+1)}{2}\nabla^2 f_{k+1} + \frac{u(u+1)(u+2)}{6}\nabla^3 f_{k+1} \right] dx$$

Since $\quad u = \dfrac{x - x_{k+1}}{h}$, therefore $dx = h\,du$

And at $\quad x = x_k \quad u = -1$

at $\quad x = x_{k+1} \quad u = 0$

Substituting these values, we get

$$y_{k+1} = y_k + h\int_0^1 \left[f_k + u\nabla f_k + \frac{u(u+1)}{2}\nabla^2 f_k + \frac{u(u+1)(u+2)}{6}\nabla^3 f_k \right] du$$

$$= y_k + h\left[f_k - \frac{1}{2}\nabla f_k - \frac{5}{12}\nabla^2 f_k + \frac{3}{8}\nabla^3 f_k \right] du$$

Expressing the backward differences in terms of function values, and substituting in the above expression, we get

$$y_{k+1} = y_k + h\int_{-1}^{0} \left[f_{k+1} + u\nabla f_{k+1} + \frac{u(u+1)}{2}\nabla^2 f_{k+1} + \frac{u(u+1)(u+2)}{6}\nabla^3 f_{k+1} \right] du$$

$$= yk + h\left[f_{k+1} - \frac{1}{2}\nabla f_{k+1} - \frac{1}{12}\nabla^2 f_{k+1} - \frac{1}{24}\nabla^3 f_{k+1} \right]$$

Expressing the backward differences in terms of function values, and substituting in the above expression, we get

$$y_{k+1}^{c} = y_k + \frac{h}{24}\left[f_{k-2} - 5f_{k-1} + 19f_k + 9f_{k+1}^{p} \right] \qquad \dots (v)$$

This is called *Adams-Moulten formula* and is used as a *corrector formula*. The superscript c indicates the predicted value of $u\,k_{+1}$. This formula can be used repeatedly to improve the accuracy of the computed value of y_{k+1}.

The Eqns. (*iii*) and (*v*) constitute the *Adams-Bashforth-Moullten predictor corrector method*. This method is popularly known as *Adams-Bashforth method*.

SOLVED EXAMPLES

Example 10.31. Given $\frac{dx}{dy} = 1 + y^2$ with $y(0) = 0$, $y(0.2) = 0.2027$, $y(0.4) = 0.4228$, and $y(0.6) = 0.6841$. Compute $y(0.8)$.

Solution: Given $h = 0.2$ and $f(x, y) = 1 + y^2$. The function values are computed at the starting points. These values are tabulated along with the starting points in the following table.

i	x	y	$f(x, y) = 1 + y^2$
0	0.2	0.0000	1.0000
1	0.2	0.2027	1.0411
2	0.4	0.4228	1.1787
3	0.6	0.6841	1.4681

To obtain $y(0.8)$, we use the predictor formula (at $k = 3$) as

$$y_{0.8}^P = y_3 + \frac{h}{24}[-9f_0 + 37f_1 - 59f_2 + 55f_3]$$

$$= 0.6841 + \frac{0.2}{24}[-9 \times 1.0 + 37 \times 1.0411 - 59 \times 1.1787 + 55 \times 1.4681]$$

$$= 1.0233 \quad \text{(Correct to four decimal places)}$$

Therefore $f_{0.8}^P = f(0.8, 1.0233) = 1 + (1.0233)^2 = 2.0471$

To correct the predicted value of $y(0.8)$, we use the corrector formula as

$$y_{0.8}^c = y_3 + \frac{h}{24}[f_1 - 5f_2 + 19f_3 + 9 f_{0.8}^P]$$

$$= 0.6841 + \frac{0.2}{24}[1.0411 - 5 \times 1.1787 + 19 \times 1.4681 + 9 \times 2.0471]$$

$$= 1.0296 \quad \text{(Correct to four decimal places)}$$

The accuracy in the value of $y(0.8) = 1.0296$ can be improved by repeatedly using the corrector formula.

Example 10.32. Apply Adams-Bashforth Method, to find a solution to the difference equation $\frac{dy}{dx} = x^2(1 + y)$ at $x = 1.4$ given $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$ and $y(1.3) = 1.979$.

Solution: Here, $h = 0.1$ and $f(x, y) = x^2(1 + y)$. The function values are computed at the starting points and are as follows.

$$x_0 = 1,\ y_0 = 1 \text{ and } f(x_0, y_0) = x_0^2(1 + y_0) = 2.000$$
$$x_1 = 1.1,\ y_1 = 1.233 \text{ and } f(x_1, y_1) = x_1^2(1 + y_1) = 2.702$$
$$x_2 = 1.2,\ y_2 = 1.548 \text{ and } f(x_2, y_2) = x_2^2(1 + y_2) = 3.669$$
$$x_3 = 1.3,\ y_3 = 1.979 \text{ and } f(x_3, y_3) = x_3^2(1 + y_3) = 5.035$$

To obtain $y(1.4)$, we use the predictor formula

$$y_4^p = y_3 + \frac{h}{24}\,[-9f(x_0, y_0) + -37f(x_1, y_1) - 59f(x_2, y_2) + 55f(x_3, y_3)]$$

$$y_4^p = 1.979 + \frac{0.1}{24}\,[-9 \times 2 + 37 \times 2.702 - 59 \times 3.699 + 55 \times 5.035] = 2.573$$

To correct the predicted value of y_4, use corrector formula

$$y_4^c = y_3 + \frac{h}{24}\,[f(x_1, y_1) - 5f(x_2, y_2) + 19f(x_3, y_3) + 9f(x_4, y_4^p)]$$

For this we compute,

$$f(x_4, y_4^p) = x_4^2\,(1 + y_4^p) = (1 + 2.573) = 7.004.$$

Therefore,

$$y_4^c = 1.979 + \frac{0.1}{24}\,[2.702 - 5 \times 3.669 + 19 \times 5.035 + 9 \times 7.004]$$

$$= 2.575$$

Therefore the solution point obtained is (1.4, 2.575).

Example 10.33. Using Runge-Kutta fourth order method to find starting solution to extrapolate the solution and apply Adam's method to find a solution to the differential equation $\frac{dy}{dx} = x - y^2$ at $x = 0.4$ in the range [0, 1] using step size 0.1 for the boundary conditions $y = 1$ at $x = 0$.

Solution: Here $x_0 = 0,\ y_0 = 1,\ h = 0.1$ and $f(x_0, y_0) = x - y^2$

The formula for Runge-Kutta fourth order method in

$$y_1 = y_0 + \frac{h}{6}\,[S_1 + 2S_2 + 2S_3 + S_4] \quad \text{... (i)}$$

Where

$$S_1 = f(x_0, y_0) = x_0 - y_0^2 = 0 - 1^2 = -1$$

$$S_2 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}S_1\right) = f\left(0 + \frac{0.1}{2}, 1 + \frac{0.1}{2} \times (-1)\right)$$

$$S_2 = f(0.05, 0.95) = 0.05 - 0.095^2 = -0.8525$$

$$S_3 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}S_2\right) = f\left(0 + \frac{0.1}{2}, 1 + \frac{0.1}{2} \times (-0.8525)\right)$$

$$S_3 = f(0.05, 0.9574) = 0.05 - (0.9574)^2 = -0.867$$

$$S_4 = f(x_0 + h, y_0 + hS_3) = f(0 + 0.1, 1 + 0.1 \times 0.867)$$

$$S_4 = f(0.1, 0.9137) = 0.1 - (0.9137)^2 = -0.7341$$

Substitute S_1, S_2, S_3 and S_4 in (i), we have the value of y at $x = 0 + 0.1 = 0.1$, *i.e.*,

$$y(0.1) = 1 + \frac{0.1}{6}\,[-1 - 2 \times 0.8525 - 2 \times 0.867 - 0.7341] = 0.9117$$

Now to find the value of y at $x = 0.1 + 0.1 = 0.2$

Here $x_1 = 01$, $y_1 = 0.9117$, $h = 0.1$.

Therefore,

$$S_1 = f(x_1, y_1) = x_1 - y_1^2 = 0.1 - 0.9117^2 = -0.731$$

$$S_2 = \left(x_1 + \frac{h}{2}, y_1 + \frac{h}{2}S_1\right) = f\left(0 + \frac{0.1}{2}, 0.9117 + \frac{0.1}{2} \times (-0.731)\right)$$

$$S_2 = f(0.15, 0.8751) = 0.15 - (0.8751)^2 = -0.616$$

$$S_3 = f\left(x_1 + \frac{h}{2}, y_1 + \frac{h}{2}S_2\right) = f\left(0.1 + \frac{0.1}{2}, 0.9117 + \frac{0.1}{2} \times (-0.616)\right)$$

$$S_3 = f(0.15, 0.8809) = 0.15 - (0.8809)^2 = -0.626$$

$$S_4 = f(x_1 + h, y_1 + hS_3) = f(0.1 + 0.1, 0.9117 + 0.1 \times (-0.626))$$

$$S_4 = f(0.2, 0.8491) = 0.2 - (0.8491)^2 = -0.521$$

Thus

$$y_2 = y_1 + \frac{h}{6}[S_1 + 2S_2 + 2S_3 + S_4]$$

$$y(0.2) = 0.9117 + \frac{0.1}{6}[-0.731 - 2 \times 0.616 - 2 \times 0.626 - 0.521] = 0.8494$$

Now to find the value of y at $x = 0.2 + 0.1 = 0.3$:

Here $x_2 = 0.2$, $y_2 = 0.8494$, $h = 0.1$

$$S_1 = f(x_2, y_2) = x_2 - y_2^2 = 0.2 - 0.8494^2 = -0.521$$

$$S_2 = f\left(x_2 + \frac{h}{2}, y_2 + \frac{h}{2}S_1\right) = f\left(0.2 + \frac{0.1}{2}, 0.8494 + \frac{0.1}{2} \times (-0.521\right)$$

$$S_2 = f(0.25, 0.8233) = 0.25 - (0.8233)^2 = -0.428$$

$$S_3 = f\left(x_2 + \frac{h}{2}, y_2 + \frac{h}{2}S_2\right) = f\left(0.2 + \frac{0.1}{2}, 0.8494 + \frac{0.1}{2} \times (-0.428\right)$$

$$S_3 = f(0.25, 0.828) = 0.25 - (0.828)^2 = -0.436$$

$$S_4 = f(x_2 + h, y_2 + hS_3) = f(0.2 + 0.1, 0.8494 + 0.1 \times (-0.436))$$

$$S_4 = f(0.3, 0.8058) = 0.25 - (0.828)^2 = -0.436$$

Thus

$$y_3 = y_2 + \frac{h}{6}[S_1 + 2S_2 + 2S_3 + S_4]$$

$$y(0.3) = 0.8494 + \frac{0.1}{6}[-0.521 - 2 \times 0.428 - 2 \times 0.436 - 0.349] = 0.8061$$

Here $h = 0.1$ and $f(x, y) = x - y^2$. The function values are computed at the starting points.

$$x_0 = 0, y_0 = 1 \text{ and } f(x_0, y_0) = x_0 - y_0^2 = 0 - 1^2 = -1$$

$$x_1 = 0.1, y_1 = 0.9117 \text{ and } f(x_1, y_1) = x_1 - y_1^2 = 0.1 - (0.9117)^2 = -0.7312$$

$$x_2 = 0.2, y_2 = 0.8494 \text{ and } f(x_2, y_2) = x_2 - y_2^2 = 0.2 - (0.8494)^2 = -0.5215$$

$$x_3 = 0.3, y_3 = 0.8061 \text{ and } f(x_3, y_3) = x_3 - y_3^2 = 0.3 - (0.8061)^2 = -0.3498$$

To obtain $y(0.4)$, we use the predictor formula

$$y_4^p = y_3 + \frac{h}{24}[-9f(x_0, y_0) + -37f(x_1, y_1) - 59f(x_2, y_2) + 55f(x_3, y_3)]$$

$$y_4^p = 0.8061 + \frac{0.1}{24}[-9 \times (-1) + 37 \times (-0.7312) - 59 \times (-0.5215) + 55 \times (-0.3498)] = 0.7789$$

To correct the predicted value of y_4, use corrector formula

$$y_4^c = y_3 + \frac{h}{24}\,[f(x_1, y_1) - 5f(x_2, y_2) + 19f(x_3, y_3) + 9f(x_4, y_4^p)]$$

For this we compute,

$$f(x_4, y_4^p) = x_4 (y_4^p)^2 = 0.4 - (0.7789)^2 = -0.2067.$$

Therefore, $y_4^c = 0.8061 + \frac{0.1}{24}\,[-0.7312 - 5 \times (-0.5215) + 19 \times (-0.3498) + 9 \times (-0.2067)]$

$$= 0.7785$$

Therefore the solution point obtained is (0.4, 0.7785).

10.8.2 Milne's Method

Milne's method, like Adams-Bashforth method, uses the information at past four solution points to extrapolate the solution at the next point. Therefore, in order to apply Milne's method three more solution points, in addition to the starting solution point, are computed using any of the methods discussed in the preceding sections.

Expression for Milne's method. Rewrite the first order ordinary differential equation

$$\frac{dy}{dx} = f(x, y) \text{ as } dy = f(x, y)\,dx$$

Integrating this equation between $x = x_{k-3}$ and $x = x_{k+1}$, we get

$$\int_{x_{k-3}}^{x_{k+1}} = \int_{x_{k-3}}^{x_{k+1}} f(x, y)\,dx$$

$$\Rightarrow \qquad y_{k+1} = y_{k+3} + \int_{x_{k-3}}^{x_{k+1}} f(x, y)\,dx \qquad \text{... (i)}$$

To solve the integral on the right, we approximate $f(x, y)$ as a polynomial in x. We derive this by making it a fit through the past points (x_{k-3}, y_{k-3}), (x_{k-2}, y_{k-2}), (x_{k-1}, y_{k-1}) and (x_k, y_k). Suppose we approximate it using Newton's forward difference interpolating polynomial as

$$f(x, y) = f_{k-3} + u\Delta f_{k-3} + \frac{u(u-1)}{2}\,\Delta^2 f_{k-3} + \frac{u(u-1)(u-2)}{6}\,\Delta^3 f_{k-3} \qquad \text{... (ii)}$$

where $u = \frac{x - x_{k-3}}{h}$, and Δf_{k-3}, $\Delta^2 f_{k-3}$, $\Delta^3 f_{k-3}$ are forward difference at point $x = x_{k-3}$.

Therefore,

$$y_{k+1} = y_{k+3} + \int_{x_{k-3}}^{x_{k+1}} \left[f_{k-3} + u\Delta f_{k-3} + \frac{u(u-1)}{2}\Delta^2 f_{k-3} + \frac{u(u-1)(u-2)}{6}\Delta^3 f_{k-3} \right] dx$$

Since $\qquad u = \frac{x - x_{k-3}}{h}$, therefore, $dx = h\,du$

And $\qquad$ at $x = x_{k-3}$ $\quad u = 0$ and at $x = x_{k+1}$ $\quad u = 4.$

To correct the predicted value of y_4, use corrector formula

$$y_4^c = y_3 + \frac{h}{24}\,[f(x_1, y_1) - 5f(x_2, y_2) + 19f(x_3, y_3) + 9f(x_4, y_4^p)]$$

For this we compute,

$$f(x_4, y_4^p) = x_4\,(y_4^p)^2 = 0.4 - (0.7789)^2 = -\,0.2067.$$

Therefore, $y_4^c = 0.8061 + \dfrac{0.1}{24}\,[-\,0.7312 - 5 \times (-\,0.5215) + 19 \times (-0.3498) + 9 \times (-\,0.2067)]$

$$= 0.7785$$

Therefore the solution point obtained is (0.4, 0.7785).

10.8.2 Milne's Method

Milne's method, like Adams-Bashforth method, uses the information at past four solution points to extrapolate the solution at the next point. Therefore, in order to apply Milne's method three more solution points, in addition to the starting solution point, are computed using any of the methods discussed in the preceding sections.

Expression for Milne's method. Rewrite the first order ordinary differential equation

$$\frac{dy}{dx} = f(x, y) \text{ as } dy = f(x, y)\,dx$$

Integrating this equation between $x = x_{k-3}$ and $x = x_{k+1}$, we get

$$\int_{x_{k-3}}^{x_{k+1}} = \int_{x_{k-3}}^{x_{k+1}} f(x, y)\,dx$$

$$\Rightarrow \qquad y_{k+1} = y_{k+3} + \int_{x_{k-3}}^{x_{k+1}} f(x, y)\,dx \qquad \text{...(i)}$$

To solve the integral on the right, we approximate $f(x, y)$ as a polynomial in x. We derive this by making it a fit through the past points (x_{k-3}, y_{k-3}), (x_{k-2}, y_{k-2}), (x_{k-1}, y_{k-1}) and (x_k, y_k). Suppose we approximate it using Newton's forward difference interpolating polynomial as

$$f(x, y) = f_{k-3} + u\Delta f_{k-3} + \frac{u(u-1)}{2}\,\Delta^2 f_{k-3} + \frac{u(u-1)(u-2)}{6}\,\Delta^3 f_{k-3} \qquad \text{...(ii)}$$

where $u = \dfrac{x - x_{k-3}}{h}$, and Δf_{k-3}, $\Delta^2 f_{k-3}$, $\Delta^3 f_{k-3}$ are forward difference at point $x = x_{k-3}$.

Therefore,

$$y_{k+1} = y_{k+3} + \int_{x_{k-3}}^{x_{k+1}} \left[f_{k-3} + u\Delta f_{k-3} + \frac{u(u-1)}{2}\Delta^2 f_{k-3} + \frac{u(u-1)(u-2)}{6}\Delta^3 f_{k-3} \right] dx$$

Since $\qquad u = \dfrac{x - x_{k-3}}{h}$, therefore, $dx = h\,du$

And $\qquad$ at $x = x_{k-3}$ $\quad u = 0$ and at $x = x_{k+1}$ $\quad u = 4.$

Substituting these values, we get

$$y_{k+1} = y_{k+3} + h\int_0^4 \left[f_{k-3} + u\Delta f_{k-3} + \frac{u(u-1)}{2}\Delta^2 f_{k-3} + \frac{u(u-1)(u-2)}{6}\Delta^3 f_{k-3} \right] du$$

$$= y_{k-3} + h\left[4f_{k-3} + 8\Delta f_{k-3} + \frac{20}{3}\Delta^2 f_{k-3} + \frac{8}{3}\Delta^3 f_{k-3} \right]$$

Expressing the forward differences in terms of function values, and substituting in the above expression, we get

$$y_{k+1}^p = y_{k-3} + \frac{4h}{3}\,[2f_{k-2} - f_{k-1} + 2f_k] \qquad \text{... (iii)}$$

This is called *Milne's formula* and is used as a *predictor formula*. The superscript p indicates the predicted value of y_{k+1}.

Now to derive the corrector formula, we construct Newton's forward differences interpolating polynomial passing through the points (x_{k-1}, y_{k-1}), (x_k, y) and (x_{k+1}, y_{k+1}^p) as

$$f(x, y) = f_{k+1} + u\Delta f_{k-1} + \frac{u(u-1)}{2}\Delta^2 f_{k-1} \qquad \text{... (iv)}$$

to evaluate the integral $\int_{x_{k-1}}^{x_{k+1}} f(x, y)\, dx$ in order to obtain the value of y_{k+1} as

$$y_{k+1} = y_{k-1} + \int_{x_{k-1}}^{x_{k+1}} f(x, y)\, dx$$

According to Simpson's 1/3rd rule for numerical integration

$$\int_{x_{k-1}}^{x_{k+1}} f(x, y)\, dx = \frac{h}{3}\,[f_{k-1} + 4f_k + f_{k+1}^p]$$

Therefore $$y_{k+1}^c = y_{k-1} + \frac{h}{3}\,[f_{k-1} + 4f_k + f_{k+1}^p] \qquad \text{... (v)}$$

This is called *Simpson's formula* and is used as a *corrector formula*. The superscript c indicates the predicted value of y_{k+1}. This formula can be used repeatedly to improve the accuracy of the computed value of y_{k+1}.

(*iii*) and (*v*) constitute the *Milne-Simpson predictor corrector method*. This method is popularly known as *Milne's method*.

SOLVED EXAMPLES

ιple 10.34. Given $\frac{dx}{dy} = 1 + y^2$ with $y(0) = 0$, $y(0.2) = 0.2027$, $y(0.4) = 0.4228$, and $y(0.6) = 0.6841$. Compute $y(0.8)$.

Solution: Given $h = 0.2$ and $f(x, y) = 1 + y^2$. The function values are computed at the starting points. These values are tabulated along with the starting points in the following table.

$$y_{k+1} = y_{k-1} + \int_{x_{k-1}}^{x_{k+1}} f(x, y)\, dx$$

According to Simpson's 1/3rd rule for numerical integration

$$\int_{x_{k-1}}^{x_{k+1}} f(x, y)\, dx = \frac{h}{3}\,[f_{k-1} + 4f_k + f_{k+1}^{p}]$$

Therefore
$$y_{k+1}^{c} = y_{k-1} + \frac{h}{3}\,[f_{k-1} + 4f_k + f_{k+1}^{p}] \qquad \text{... }(v)$$

This is called *Simpson's formula* and is used as a *corrector formula*. The superscript c indicates the predicted value of y_{k+1}. This formula can be used repeatedly to improve the accuracy of the computed value of y_{k+1}.

(*iii*) and (*v*) constitute the *Milne-Simpson predictor corrector method*. This method is popularly known as *Milne's method*.

SOLVED EXAMPLES

Example 10.34. **Given $\frac{dx}{dy} = 1 + y^2$ with $y(0) = 0$, $y(0.2) = 0.2027$, $y(0.4) = 0.4228$, and $y(0.6) = 0.6841$. Compute $y(0.8)$.**

Solution: Given $h = 0.2$ and $f(x, y) = 1 + y^2$. The function values are computed at the starting points. These values are tabulated along with the starting points in the following table.

i	x	y	$f(x, y) = 1 + y^2$
0	0.0	0.0000	1.0000
1	0.2	0.2027	1.0411
2	0.4	0.4228	1.1787
3	0.6	0.6841	1.4681

To obtain $y(0.8)$, we use the predictor formula (at $k = 3$) as

$$y^p_{0.8} = y_0 + \frac{4h}{3}[2f_1 - f_2 + 2f_3]$$

$$= 0.0 + \frac{4 \times 0.2}{3}[2 \times 1.0411 - 1.1787 + 2 \times 1.4681]$$

$$= 1.0239 \quad \text{(Correct to four decimal places)}$$

Therefore $f^p_{0.8} = f(0.8, 1.0239) = 1 + (1.0239)^2 = 2.0484$

To correct the predicted value of $y(0.8)$, we use the corrector formula as

$$y^c_{0.8} = y_2 + \frac{h}{3}[f_2 - 4f_3 + f^p_{0.8}]$$

$$= 0.4228 + \frac{0.2}{3}[1.1787 + 4 \times 1.4681 + 2.0484]$$

$$= 1.0294 \quad \text{(Correct to four decimal places)}$$

The accuracy in the value of $y(0.8) = 1.0296$ can be improved by repeatedly using the corrector formula.

Example 10.35. **Using Picard's method to find first four values to extrapolate the solution and apply Milne's method to find a solution to the differential equation $\frac{dy}{dx} = x - y^2$ in the range [0, 1] using step size 0.2 for the boundary conditions $y = 0$ at $x = 0$.**

Solution: To find first four prior values to extrapolate the solution at the next point using Picard's method. Since $f(x, y) = x = y^2$ and $y = 0$ when $x_0 = 0$, we have integral equation

$$y = y_0 + \int_{x_0}^{x} f(x, y)\, dx$$

becomes

$$y = 0 + \int_{x_0}^{x} f(x - y^2)\, dx \quad \text{... (i)}$$

First approximation:

Put $y = 0$ in $(x - y^2)$ of (1), we get,

$$y_1 = 0 + \int_{x_0}^{x} x\, dx = \frac{x^2}{2}$$

Second approximation:
Put

$$y = x^2/2 \text{ in } (x - y^2) \text{ of (1), we get,}$$

$$y_2 = 0 + \int_{x_0}^{x}\left(x - \frac{x^4}{4}\right) dx = \frac{x^2}{2} - \frac{x^5}{20}$$

Third approximation:

Put

$$y = \left(\frac{x^2}{2} - \frac{x^5}{20}\right) \text{ in } (x - y^2) \text{ of (1), we get,}$$

$$y_3 = 0 + \int_{x_0}^{x}\left(x - \left(\frac{x^2}{2} - \frac{x^5}{20}\right)^2\right) dx = \frac{x^2}{2} - \frac{x^5}{20} + \frac{x^8}{160} - \frac{x^{11}}{4400}$$

Compute the first four values and their function values for the Milne's method.

$$x_0 = 0, \quad y_0 = 0 \quad \text{and} \quad f(x_0, y_0) = x_0 - y_0^2 = 0$$

$$x_1 = 0.2, \quad y_1 = \frac{x_1^2}{2} = 0.02$$

and $$f(x_1, y_1) = x_1 - y_1^2 = 0.1996$$

$$x_2 = 0.4, \quad y_2 = \left(\frac{x_2^2}{2} - \frac{x_2^5}{20}\right) = 0.0795$$

and $$f(x_2, y_2) = x_2 - y_2^2 = 0.3937$$

$$x_3 = 0.6, \quad y_3 = \left(\frac{x_3^2}{2} - \frac{x_3^5}{20}\right) + \left(\frac{x_3^8}{160} - \frac{x_3}{4400}\right) = 0.1762$$

and $$f(x_3, y_3) = x_3 - y_3^2 = 0.5689$$

To obtain $y(0.8)$, we use Milne's predictor formula

$$y_4^p = y_0 + \frac{4h}{3}\left[2f(x_1, y_1)\, f(x_2, y_2) + 2f(x_3, y_3)\right)$$

$$y_4^p = 0 + \frac{0.8}{3}\left[2 \times 0.1996 - 0.3937 + 2 \times 0.5689\right] = 0.3049$$

and $$x_4 = x_3 + h = 0.6 + 0.2 = 0.8$$

To correct the predicted value of y_4, use Milne's corrector formula

$$y_4^c = y_2 + \frac{h}{3}\left[f(x_2, y_2) + 4f(x_3, y_3) + f(x_4, y_4^p)\right]$$

For this we compute, $f(x_4, y_4^p) = x_4\,(y_4^p) = 0.8 - (0.3049)^2 = 0.7070$

$$y_4^c = 0.0795 + \frac{0.2}{3}\left[0.3937 + 4 \times 0.5689 + 0.7070\right] = 0.3046$$

Now again using the predictor formula by taking the solution point (0.8, 0.3046)

$$y_5^p = y_0 + \frac{4h}{3}\left[2f(x_2, y_2) - f(x_3, y_3) + 2f(x_4, y_4)\right]$$

For this we compute, $f(x_4, y_4) = x_4 (y_4)^2 = 0.8 - (0.3046)^2 = 0.7072.$

Therefore, we get

$$y_5^p = 0.02 + \frac{0.8}{3}[2 \times 0.3937 - 0.5689 + 2 \times 0.7072] = 0.4554$$

and $$x_5 = x_4 + h = 0.8 + 0.2 = 1.0$$

To correct the predicted value of y_4, use Milne's corrector formula

$$y_5^c = y_3 + \frac{h}{3}[f(x_3, y_3) + 4f(x_4, y_4) + f(x_5, y_5^p)]$$

For this we compute, $f(x_5, y_5^p) = x_5 - \left(y_5^p\right)^2 = 1.0 - (0.4554)^2 = 0.7926.$

Therefore, we get,

$$y_5^c = 0.1762 + \frac{0.2}{3}[0.5689 + 4 \times 0.7072 + 0.7926] = 0.4555$$

Thus the solution point obtained is (1.0, 0.4555).

Example 10.36. Apply Milne's method, to find a solution to the differential equation $\frac{dy}{dx} = 1 + xy^2$ as $x = 0.8$, given $y(0) = 0$, $y(0.2) = 0.2027$, $y(0.4) = 0.4228$ and $y(0.6) = 0.6841$.

Solution: Here $h = 0.2$ and $f(x, y) = 1 + xy^2$. The function values are computed at the starting points, *i.e.*,

$$x_0 = 0, \quad y_0 = 0 \quad \text{and} \quad f(x_0, y_0) = 1 + x_0 y_0^2 = 1$$
$$x_1 = 0.2, \quad y_1 = 0.2027 \quad \text{and} \quad f(x_1, y_1) = 1 + x_1 y_1^2 = 1.0082$$
$$x_2 = 0.4, \quad y_2 = 0.4228 \quad \text{and} \quad f(x_2, y_2) = 1 + x_2 y_2^2 = 1.0715$$
$$x_3 = 0.6, \quad y_3 = 0.6841 \quad \text{and} \quad f(x_3, y_3) = 1 + x_3 y_3^2 = 1.2808$$

To obtain $y(0.8)$, we use the Milne's predictor formula

$$y_4^p = y_0 + \frac{4h}{3}[2f(x_1, y_1) - f(x_2, y_2) + 2f(x_3, y_3)]$$

$$y_4^p = 0 + \frac{4 \times 0.2}{3}[2 \times 1.0082 - 1.0715 + 2 \times 1.2808] = 0.9351$$

and $$x_4 = x_3 + h = 0.6 + 0.2 = 0.8$$

To correct the predictor value of x_4, use Milne's corrector formula

$$y_4^c = y_2 + \frac{h}{3}[f(x_2, y_2) + 4f(x_3, y_3) + f(x_4, y_4^p)]$$

For this we compute, $f(x_4, y_4^p) = 1 + x_4 (y_4^p)^2 = 1 + 0.8\,(0.9351)^2 = 1.6995$

Thus

$$y_4^c = 0.4228 + \frac{0.2}{2}[1.0715 + 4 \times 1.2808 + 1.6995] = 1.2122$$